

Week 5 Recitation

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ECON8711 (Micro 1A)

Sep 22, 2023

Please let me know your thoughts and suggestions
till every recitation class:

<https://forms.gle/jm3d4iJ51tjUnTiD9>

Profit max & Cost min

(MWG Chap 5, Section B,C)

Production function $f(x) = y$



Profit Max:

$$\max_{y \in \mathbb{R}^N} p \cdot y \text{ s.t. } F(y) \leq 0$$

(p, w_1, w_2)
 $y - f(x_1, x_2) \leq 0$
 $y \leq f(x_1, x_2)$

Cost min:

$$\min_{w \in \mathbb{R}_+^N} w \cdot z \text{ s.t. } f(z) \geq q$$

Lagrangian Practice

(In-class example)

Consider the production function ✓

$$f(x) = \underbrace{x_1^{\alpha_1}} \underbrace{x_2^{\alpha_2}} \quad \text{s.t.} \quad \alpha_1 + \alpha_2 = 1$$

→ Set up the profit max problem? monotone ⇒ constraint binding

$$\max_{\substack{(y, \kappa_1, \kappa_2) \in \mathbb{R}_+^N \\ \sim \sim \sim}}$$

$$\underbrace{g(y, \kappa_1, \kappa_2)}_{p y - w_1 \kappa_1 - w_2 \kappa_2} \quad \text{s.t.}$$

$$y \leq \kappa_1^{\alpha_1} \kappa_2^{\alpha_2} \\ \uparrow \\ \underline{y = \kappa_1^{\alpha_1} \kappa_2^{\alpha_2}}$$

$$\vec{p} = (p, w_1, w_2), \quad \vec{y} = (y, \kappa_1, \kappa_2)$$

- CRS production function

⇒ unconstrained problem:

$$\max_{\kappa_1, \kappa_2 \in \mathbb{R}_+^N}$$

$$\underbrace{p(\underbrace{\kappa_1^{\alpha_1} \kappa_2^{\alpha_2}}_g)} - w_1 \kappa_1 - w_2 \kappa_2 \quad \checkmark$$

$$\underline{\kappa_1, \kappa_2 \geq 0.}$$

$$\alpha_1 = 1 - \alpha_2$$

$$\mathcal{L} = g + \mu_1 \kappa_1 + \mu_2 \kappa_2$$

$$\mathcal{L} = p y - w_1 \kappa_1 - w_2 \kappa_2 - \lambda [y - \kappa_1^{\alpha_1} \kappa_2^{\alpha_2}] + \mu_1 \kappa_1 + \mu_2 \kappa_2 \quad \checkmark$$

$$\text{FOC: } [\alpha_1] \quad \checkmark \quad p\alpha_1 \left(\frac{x_2}{x_1} \right)^{\alpha_2} - w_1 = 0 \Rightarrow \left(\frac{x_2}{x_1} \right)^{\alpha_2} = \frac{w_1}{p\alpha_1} \dots \textcircled{1}$$

$$[\alpha_2] \quad p\alpha_2 \left(\frac{x_1}{x_2} \right)^{\alpha_1} - w_2 = 0 \Rightarrow \left(\frac{x_1}{x_2} \right)^{\alpha_1} = \frac{w_2}{p\alpha_2} \dots \textcircled{2}$$

$$\textcircled{1}: \left(\frac{x_2}{x_1} \right) = \left(\frac{w_1}{p\alpha_1} \right)^{\frac{1}{\alpha_2}}$$

$$\textcircled{2}: \frac{x_2}{x_1} = \left(\frac{w_2}{p\alpha_2} \right)^{-\frac{1}{\alpha_1}} = \left(\frac{p\alpha_2}{w_2} \right)^{\frac{1}{\alpha_1}}$$

$$\left(\frac{w_1}{p\alpha_1} \right)^{\frac{1}{\alpha_2}} = \left(\frac{p\alpha_2}{w_2} \right)^{\frac{1}{\alpha_1}} \checkmark$$

$$\left(\frac{w_1}{p\alpha_1} \right) = \left(\frac{p\alpha_2}{w_2} \right)^{\frac{1-\alpha_1}{\alpha_1}}$$

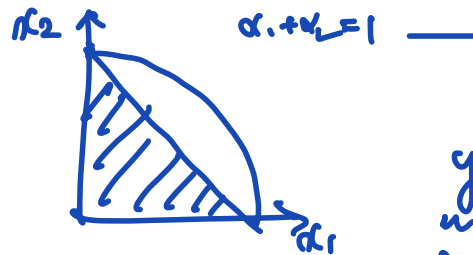
$$w_1 = \sim$$

$$w_2 = \sim$$



Lagrangian Practice

$$\underbrace{\ln(\alpha_1 + 2\alpha_2 + \alpha_3)}_{y_1} + \underbrace{\ln(\alpha_1 + 3\alpha_3)}_{y_2}$$



$$y = \min \{y_1, y_2\}$$



- Existence of an optimizer:
concavity/convexity of the objective, ~~concavity~~ non convexity / convexity of the constraint, ...
- * Monotonic objective function $\Rightarrow$ binding constraints
- Setting up Lagrangian, Complementary slackness conditions
- FOC and try to solve

Lagrangian Practice

(In-class example)

Consider the cost minimization problem

$$\min_{x \in \mathbb{R}_+^2} w_1 x_1 + w_2 x_2 \text{ s.t. } x_1^{\alpha_1} x_2^{\alpha_2} \geq \bar{y}$$

Lagrangian Practice

2022 Quiz 3, #2

$$\begin{aligned} \max_{y \in \mathbb{R}_+^2} \quad & 0.25 \log(y_1) + 0.75 \log(y_2) \quad \checkmark \\ \text{s.t.} \quad & 5y_1 + 10y_2 \leq 20 \end{aligned}$$

- (a) Set up the Lagrangian function for the optimization problem above. Be sure to include all constraints and put correct sign restrictions on the Lagrange multipliers. ✓
- ✓ (b) Are there any constraints that you can drop from the Lagrangian? Justify your answer.

$$\begin{aligned} \max_{y \in \mathbb{R}_+^2} \quad & \frac{1}{4} \ln(y_1) + \frac{3}{4} \ln(y_2) \\ \text{s.t.} \quad & 5y_1 + 10y_2 \leq 20 \end{aligned}$$

$$L = \frac{1}{4} \ln(y_1) + \frac{3}{4} \ln y_2 + \lambda [20 - 5y_1 - 10y_2] + \cancel{\mu_1 y_1} + \cancel{\mu_2 y_2}$$

$$\begin{aligned} \text{FOC} \\ \Rightarrow [y_1]: \quad & \frac{1}{4y_1} - 5\lambda + \cancel{\mu_1} = 0 \dots\dots (1) \quad \begin{array}{l} \lambda \geq 0 \\ \mu_1 \geq 0 \\ \mu_2 \geq 0 \\ \mu_1 = \mu_2 = 0 \end{array} \\ [y_2]: \quad & \frac{3}{4y_2} - 10\lambda + \cancel{\mu_2} = 0 \dots\dots (2) \\ [\lambda]: \quad & 20 - 5y_1 - 10y_2 = 0 \dots\dots (3) \end{aligned}$$

$$(1): \quad \lambda = \frac{1}{20y_1}$$

$$(2): \quad \lambda = \frac{3}{40y_2}$$

$$\lambda = \frac{1}{20y_1} = \frac{3}{40y_2}$$

$$3y_1 = 2y_2 \Leftrightarrow 10y_2 = 15y_1$$

$$20 = 5y_1 + 15y_1$$

$$y_1^* = 1$$

$$y_2^* = \frac{2}{3}$$

$$\lambda^* = 1$$

Lagrangian Practice

Fixing one variable

(c) Solve the maximization problem when the Lagrange multiplier on the budget is equal to 2. Solve the Lagrangian finding the optimal value of the Lagrange multiplier. How do the two solutions compare? Explain why they are different or the same.

$$[y_1]: \frac{1}{4y_1} - 5\lambda + \cancel{y_1^0} = 0 \dots\dots ①$$

$$[y_2]: \frac{3}{4y_2} - 10\lambda + \cancel{y_2^0} = 0 \dots\dots ②$$

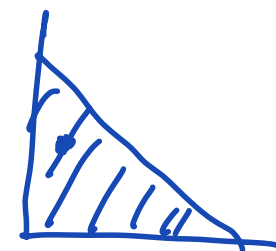
$$[\lambda]: 20 - 5y_1 - 10y_2 = 0 \dots\dots ③$$

↑

$$\begin{aligned} \text{①: } \lambda=2 & \Rightarrow \frac{1}{10} = \frac{1}{4y_1} \Rightarrow \tilde{y}_1 = \frac{1}{40} \\ \text{②: } 20 & = \frac{3}{4y_2} \Rightarrow \tilde{y}_2 = \frac{3}{80} \end{aligned}$$

$$y_1^* = 1, y_2^* = \frac{3}{2}$$

$$5\tilde{y}_1 + 10\tilde{y}_2 = \frac{1}{8} + \frac{3}{8} = \frac{1}{2} \ll 20 \text{ (budget not binding)} \leftarrow \lambda \uparrow: \text{over}$$



Lagrangian Practice

Fixing one variable

(e.g. short-run production vs. long-run production)

(In class example, revisit)

Consider the cost minimization problem

$$\min_{x \in \mathbb{R}_+^2} w_1 \overline{x}_1 + w_2 \underline{x}_2 \text{ s.t. } x_1^{\alpha_1} x_2^{\alpha_2} \geq \bar{y}$$

Now, fix $x_1 = \bar{x}_1$ ✓